

Question Paper

Exam Date & Time: 27-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER M.TECH EXAMINATIONS - NOVEMBER 2025
Computational Methods and Stochastic Process [MAT_5128]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) (4)
- A) Apply Gram Schmidt orthogonalization process to construct an orthonormal basis of $\mathbb{R}^3$ from the following linearly independent set $\{(1, 2, 2), (5, 1, 1), (-3, 4, 2)\}$.

Course Outcome: "CO1" / Cognitive Level: "Application"

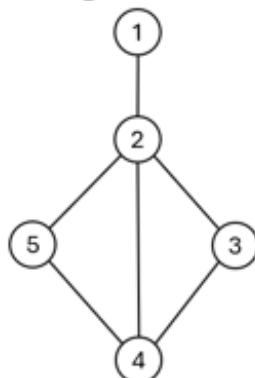
- B) (3)
- Find the QR decomposition of $A = \begin{bmatrix} 4 & 7 \\ 2 & 9 \end{bmatrix}$

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) (3)
- A company has three machines A, B, and C producing 20%, 30%, and 50% of its total output, respectively. The percentage of defective items produced by these machines are 6%, 3%, and 2%, respectively. An item is picked at random and is found to be defective. Find the probability that it was produced by machine A.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 2) (4)
- A) Using Kirchhoff's Matrix-Tree Theorem, determine the number of spanning trees of the following graph.



B) (3)

Solve $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$, $0 < x < 1, t > 0$,
 $u(x, 0) = 100x^2, \frac{\partial u}{\partial t}(x, 0) = 0$,
 $u(0, t) = u(1, t) = 0$ with $h = 0.25$ up to four steps.

C) (3)

Minimize by graphical method, $Z = 6x + 9y$
 subject to

$$\begin{aligned} x + y &\geq 8, \\ 2x + y &\geq 10, \\ x &\geq 0, y \geq 0. \end{aligned}$$

3) (4)

Maximize by Simplex method, $Z = 2x_1 + 3x_2$
 subject to

A)

$$\begin{aligned} x_1 + x_2 &\leq 4, \\ x_1 + 3x_2 &\leq 6, \\ x_1, x_2 &\geq 0. \end{aligned}$$

B) (3)

Use the dominance method to reduce the following game between player A and player B and determine its solution.

	Player B			
Player A	3	2	4	0
	1	3	2	1
	4	3	5	2

C) (3)

Two cafés A and B open across the street.

Café A can choose Premium Price, Regular Price, or Discount Price.

Café B can choose No Promotion, Small Promotion, or Large Promotion.

- If A chooses Premium and B does No Promotion, A gains ₹6k market advantage; with Small Promotion A gains ₹3k; with Large Promotion A loses ₹2k.
- If A chooses Regular and B does No Promotion A gains ₹4k; with Small Promotion ₹4k; with Large Promotion ₹1k.
- If A chooses Discount and B does No Promotion A gains ₹1k; with Small Promotion A gains ₹2k; with Large Promotion A loses ₹3k.

Construct the payoff matrix for A and determine whether a saddle point exists. If yes, state the equilibrium and value.

- 4) Two different training programs (X and Y) are introduced to improve typing speed (in words per minute). (4)
- A)

A random sample of 6 trainees is selected for each program.

Program X	65	70	67	71	66	69
Program Y	60	56	63	58	62	61

Test whether there is a significant difference in the mean accuracies of the two programs at $\alpha = 0.01$. Assume the two samples are independent and have equal variances.

- B) A factory produces items using two machines, M_1 and M_2 . Machine M_1 produces 60% of the total items, and M_2 produces 40%. (3)

Out of the items produced:

- 5% of M_1 's items are defective.
- 10% of M_2 's items are defective.

Let X be Machine used (M_1 or M_2) and Y be Condition of item (Defective or Non-defective).

Construct the joint probability distribution table of X and Y .

- C) The four independent normal random variables X_1, X_2, X_3 and X_4 have the following means and variances. (3)

	Mean	Variance
X_1	13	5
X_2	17	7
X_3	11	3
X_4	19	9

Find mean, variance and standard deviation of

$$Q = 2X_1 + 3X_2 - 5X_3 + 7X_4 + 11.$$

- 5) (4)
- A)

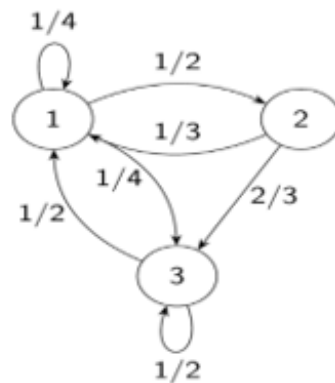
A basket contains 4 apples and 2 oranges. The following experiment is repeated indefinitely: A fruit is chosen at random from the basket. If the fruit is an apple, it is put back into the basket. If the fruit is an orange, it is not replaced. A coin is selected at random from the box. If the coin is silver, it is put back into the box. If the coin is gold, it is not replaced. Let X_n denote the number of oranges remaining in the basket after n selections. Find the transition probability matrix of the Markov chain $\{X_n\}$.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

B)

Transition diagram is as follows.

(3)



If we know $P(X_0 = 1) = \frac{1}{3}$, find

- (i) Transition probability matrix.
- (ii) $P(X_0 = 1, X_1 = 2)$
- (iii) $P(X_0 = 1, X_1 = 2, X_2 = 3)$

Course Outcome: "CO5" / Cognitive Level: "Application"

C)

A company experiences either High Demand (H) or Low Demand (L) for its product each month. If the demand is high this month, the probability it will stay high next month is 0.75. If the demand is low this month, the probability it becomes high next month is 0.4. Find the transition probability matrix and compute the long-run probabilities of High and Low demand.

(3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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#	Course Outcome	Description	Marks
1	1	CO 1	7
2	2	CO 2	7
3	3	CO 3	13
4	4	CO 4	13
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025-PG MAT_5128:Computational Methods and Stochastic Process

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	8	26(52%)
4	Analysis	7	24(48%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)